

ABSTRACT

In the rapidly expanding gaming industry, users often face difficulty discovering games that suit their preferences. To address this, our project presents a game recommendation system that combines advanced recommendation strategies to provide personalized suggestions. The system integrates content-based filtering, collaborative filtering, and a hybrid filtering mechanism to enhance recommendation accuracy. For feature representation, we use both TF-IDF for textual analysis and BERT-based sentence embeddings for semantic understanding of game descriptions. Collaborative filtering leverages user interaction data to identify shared behavioral patterns, while the hybrid model merges both approaches to improve relevance and diversity of recommendations. The system also includes tailored strategies to address cold-start problems, ensuring meaningful suggestions even for new users without prior interaction history. The project is built using Python for backend logic and Next.js (TypeScript) for the frontend. This system highlights the potential of intelligent recommendation strategies to deliver smart, scalable, and engaging game suggestions.